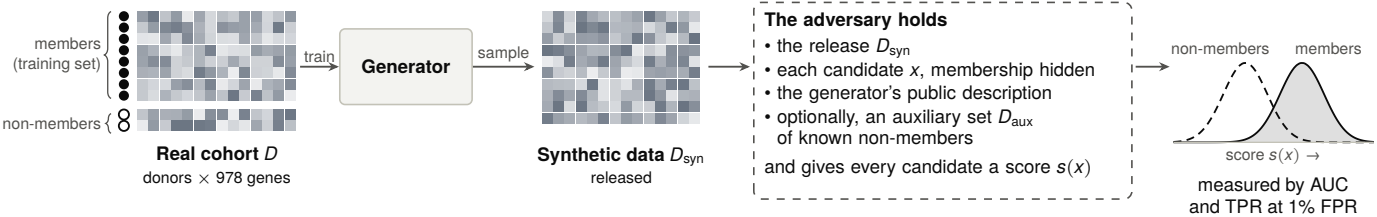
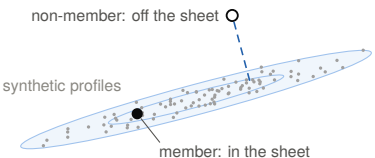
a The membership game



b Three attacks, each reading the place where a generator family keeps its training donors

MahalaMIA — reads the covariance

against MVN and the CVAE



$s(x)$ is minus the Mahalanobis distance from x to the synthetic profiles of its class.

MeLoMIA — reads the losses

against CVAE, NoisyDiffusion, TabSyn, TabPFN

repeated K times, on known splits of D:

shadow generator → samples → synth-shadow h_k → losses → classifier $\mathcal{M}$

at attack time:

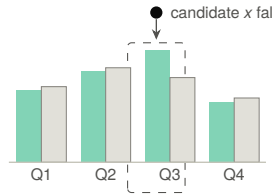
release D_{syn} → train → proxy h^* → losses → classifier $\mathcal{M}$

trained on synthetic data only, so $\mathcal{M}$ learns from h_k and scores with h^*, the same kind of model

Each loss is calibrated against the same profile's losses under the K synth-shadows.

MAMA-MIA — reads the measured marginals

against DP graphical models



$s(x) = \sum_t \log \hat{p}^t_{syn}(x_t) / \hat{p}^t_{cand}(x_t)$ over the measured tables t: a member added one count to its cell of each.